



Ravenswood

16 June 2016

Assessment 3/ HSC Assessment 2 Conics and Integration

4 Copies

Year 12

Mrs Kim

Mathematics Extension 2

PART I: CONICS

General Instructions

- ◆ Working time **40 minutes**
- ◆ Write using blue or black pen
- ◆ Draw diagrams in pencil
- ◆ Board-approved calculators may be used
- ◆ All necessary working should be shown in every question
- ◆ Marks will be deducted for careless or badly arranged work.

Total marks (28)

- ◆ Use the Multiple Choice answer sheet provided for Section I
- ◆ Answer the questions in answer booklets marked with your student number and the Question Number for Section II
- ◆ If you have not attempted a question or part of a question, write the question number with "NO ATTEMPT" beside it.

Section I Multiple Choice

There are 4 questions in this section. Use the multiple choice sheet provided to record your answers.

1. What are the coordinates of the foci of the ellipse with equation $\frac{x^2}{64} + \frac{y^2}{36} = 1$?
 - (A) $(-10, 0)$ and $(10, 0)$
 - (B) $(-2\sqrt{7}, 0)$ and $(2\sqrt{7}, 0)$
 - (C) $(0, -2\sqrt{7})$ and $(0, 2\sqrt{7})$
 - (D) $(-28, 0)$ and $(28, 0)$

2. What is the equation of the directrices of the hyperbola with parametric coordinates $(10 \sec \theta, 5 \tan \theta)$?
 - (A) $x = \pm 5\sqrt{3}$
 - (B) $x = \pm 5\sqrt{5}$
 - (C) $x = \pm 4\sqrt{3}$
 - (D) $x = \pm 4\sqrt{5}$

3. The ellipse with a focus at $(4, 0)$ and directrix $x = 8$ has equation:
 - (A) $\frac{x^2}{16} + \frac{y^2}{32} = 1$
 - (B) $\frac{x^2}{32} + \frac{y^2}{16} = 1$
 - (C) $\frac{x^2}{16} - \frac{y^2}{32} = 1$
 - (D) $\frac{x^2}{32} - \frac{y^2}{16} = 1$

4. What is the equation of the chord of contact from the point (x_0, y_0) to the hyperbola

$$\frac{x^2}{10} - \frac{y^2}{5} = 1?$$

(A) $\frac{xx_0}{10} - \frac{yy_0}{5} = 1$

(B) $\frac{xx_0}{5} - \frac{2yy_0}{5} = 1$

(C) $\frac{x_0^2}{10} - \frac{y_0^2}{5} = 1$

(D) $\frac{x_0 \sec \theta}{10} - \frac{y_0 \tan \theta}{5} = 1$

End of Section I

Section II.

There are 2 questions in this section.

Complete your solutions in the booklets provided. Please start each question in a new booklet.

Question 5 (12 Marks)

- (a) Draw a neat sketch of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$. 3

On your diagram show the coordinates of the foci, the equations of the directrices and asymptotes.

- (b) A hyperbola has the equation: $\frac{x^2}{3} - \frac{y^2}{2} = 1$.

- (i) Find the coordinates of the foci. 2

- (ii) The equation of the tangent to the hyperbola at $P(\sqrt{3} \sec \theta, \sqrt{2} \tan \theta)$ is:

$$\sqrt{2}x \sec \theta - \sqrt{3}y \tan \theta = \sqrt{6}.$$

(Do not prove this equation)

Show that the equation of the normal at P is: 2

$$\sqrt{3}x \tan \theta + \sqrt{2}y \sec \theta = 5 \sec \theta \tan \theta.$$

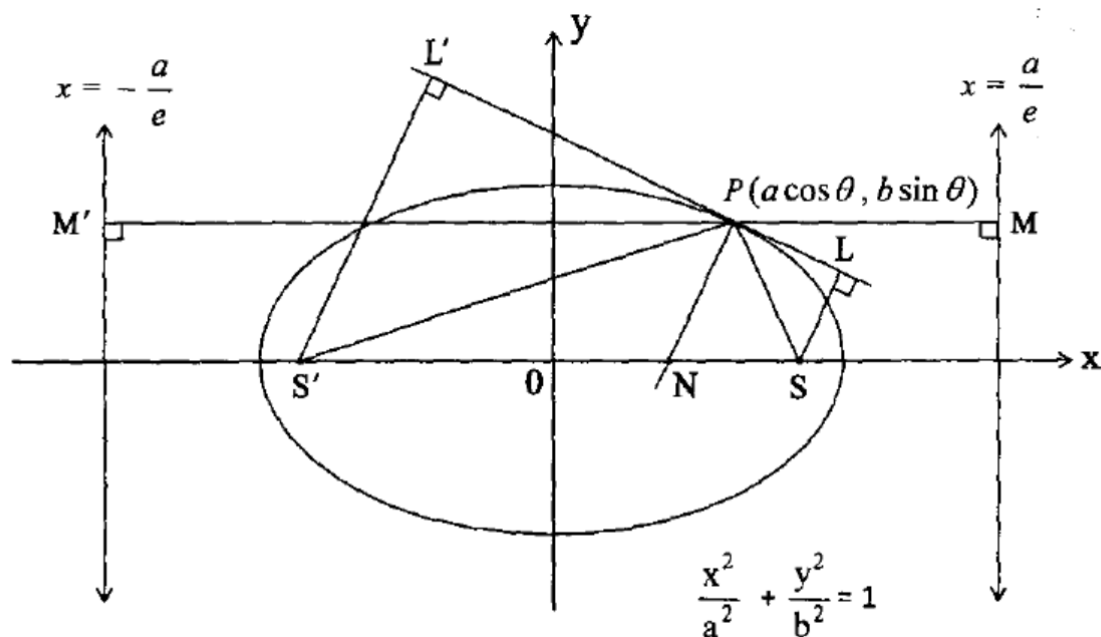
- (iii) The tangent and normal to the hyperbola at P cut the y axis at T and N respectively.

Show that the circle with TN as diameter passes through the foci of the hyperbola. 5

Question 6 (12 Marks)

Lines drawn from the foci S and S' of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, are perpendicular to the tangent drawn at $P(a\cos\theta, b\sin\theta)$. They meet this tangent at L and L' respectively.

The line parallel to the x – axis passing through P intersects the directrices at M and M' and the normal at P meets the x –axis at N .



- | | | |
|-------|--|---|
| (i) | Show that $PS = a(1 - e\cos\theta)$ | 2 |
| (ii) | Write down a similar expression for PS' | 1 |
| (iii) | Show that the equation of the tangent at P is $bx\cos\theta + ay\sin\theta - ab = 0$. | 2 |
| (iv) | Find the distances SL and $S'L'$ from the foci S and S' to the tangent at P . | 2 |
| (v) | Hence, or otherwise, show that PN bisects $\angle SPS'$. | 3 |
| (vi) | Show that $\frac{PS}{NS} = \frac{PS'}{NS'}$ | 2 |

End of Section II



--	--	--	--	--	--	--	--

MULTIPLE CHOICE ANSWER SHEET

Section I – Questions 1 – 4

Sample

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

$2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word correct and drawing an arrow as follows:

A ☒ B ☒ ^{correct} C ☐ D ☐

SECTION 1: Colour in the appropriate circle.

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐



Ravenswood

16 June 2016

Assessment 3/ HSC Assessment 2 Conics and Integration

4 Copies

Year 12

Mrs Kim

Mathematics Extension 2

PART II: Integration

General Instructions

- ◆ Working time **40 minutes**
- ◆ Write using blue or black pen
- ◆ Draw diagrams in pencil
- ◆ Board-approved calculators may be used
- ◆ All necessary working should be shown in every question
- ◆ Marks will be deducted for careless or badly arranged work.

Total marks (24)

- ◆ Use the Multiple Choice answer sheet provided for Section I
- ◆ Answer the questions in answer booklets marked with your student number and the Question Number for Section II
- ◆ If you have not attempted a question or part of a question, write the question number with “NO ATTEMPT” beside it.

Section I Multiple Choice

There are 4 questions in this section. Use the multiple choice sheet provided to record your answers.

1. To solve $\int_{\sqrt{3}}^2 \sqrt{4-x^2}$, which of the following methods would you apply?

- (A) Integration by parts
- (B) Partial fractions
- (C) Substitution with $x = \cos \theta$
- (D) Substitution with $x = 2 \sin \theta$

2. The integral of $x^2 e^x + 2x e^x$ is:

- (A) $x^2 + e^x + c$
- (B) $x^2 e^x + c$
- (C) $x e^x + c$
- (D) $2x e^x + c$

3. Which of the following is an expression for $\int \frac{dx}{\sqrt{7-6x-x^2}}$?

- (A) $\sin^{-1}\left(\frac{x-3}{2}\right) + c$
- (B) $\sin^{-1}\left(\frac{x+3}{2}\right) + c$
- (C) $\sin^{-1}\left(\frac{x-3}{4}\right) + c$
- (D) $\sin^{-1}\left(\frac{x+3}{4}\right) + c$

4. If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ where n is a positive integer, and $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$ for $n \geq 2$, then $I_n = \frac{n-1}{n} I_{n-2}$, for $n \geq 2$. What is the value of I_4 ?

(A) $\frac{\pi}{16}$

(B) $\frac{3\pi}{16}$

(C) $\frac{3}{8}$

(D) $\frac{3\pi}{8}$

End of Section I

Section II.

There are 2 questions in this section.

Complete your solutions in the booklets provided. Please start each question in a new booklet.

Question 5 (10 Marks)

(a) Find $\int \frac{x^2}{\sqrt{x^3-1}} dx$ 2

(b) Evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x dx$ 3

(c) (i) Find A , B and C if $\frac{x^2-4x+2}{(2x+1)(x^2+4)} = \frac{A}{2x+1} + \frac{Bx+C}{x^2+4}$ 3

(ii) Hence, evaluate $\int_0^2 \frac{x^2-4x+2}{(2x+1)(x^2+4)} dx$. 2

Leave your answer in simplest exact form.

Question 6 (10 Marks)

(a) Let $t = \tan \frac{\theta}{2}$

(i) Show that $d\theta = \frac{2}{1+t^2} dt$ 2

(ii) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 2\operatorname{cosec}^2 \theta \tan \frac{\theta}{2} d\theta$ 3

Question 6 continues on the following page.

Question 6 (continued)

- (b) (i) Given that $I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$ where n is a positive integer, **2**
show that $I_{2n+1} = \frac{1}{2} e - n I_{2n-1}$.
- (ii) Hence, or otherwise, evaluate $\int_0^1 x^5 e^{x^2} dx$. **3**

End of Section II



--	--	--	--	--	--	--	--

Student Number

MULTIPLE CHOICE ANSWER SHEET

Section I – Questions 1 – 4

Sample

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

$2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word correct and drawing an arrow as follows:

A ☒ B ☒ ^{correct} C ☐ D ☐

SECTION 1: Colour in the appropriate circle.

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

CONICS - Please Learn!Ellipse

- $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
- $a^2 = b^2(1 - e^2)$
- $e = \sqrt{1 - \frac{b^2}{a^2}}$
- Focus: $(\pm ae, 0)$
- Equation of Directrix: $x = \pm \frac{a}{e}$
- Tangent Equation: $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
- Normal equation: $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$
- chord of contact: $\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = 1$

Hyperbola

- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- asymptotes: $y = \pm \frac{bx}{a}$
- $b^2 = a^2(e^2 - 1)$
- $e = \sqrt{1 + \frac{b^2}{a^2}}$
- focus: $(\pm ae, 0)$
- directrix: $x = \pm \frac{a}{e}$
- Tangent equation: $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$
- Normal equation: $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$
- chord of contact: $\frac{x_0x}{a^2} - \frac{y_0y}{b^2} = 1$

Section I (MC)

Q1. $S = (\pm ae, 0)$.

$a = 8, b = 6$

$\therefore S(\pm 8e, 0)$.

$$e = \sqrt{1 - \frac{36}{64}}$$

$$= \frac{2\sqrt{7}}{8}$$

$$\therefore S(\pm 2\sqrt{7}, 0) \Rightarrow \text{Answer } \boxed{B}$$

Q2. $a = 10, b = 5$.

$$x = \pm \frac{a}{e}$$

$$= \pm \frac{10}{e}$$

$$e = \sqrt{1 + \frac{25}{100}}$$

$$= \frac{\sqrt{5}}{2}$$

$$\therefore x = \pm \frac{20}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$= \pm 4\sqrt{5} \Rightarrow \boxed{D}$$

Q3. Focus $(4, 0)$.
directrix $x = 8$.

$\therefore 4 = ae$

$x \pm 8 = \frac{a}{e}$

$$e = \sqrt{1 - \frac{16}{32}}$$

$$= \sqrt{\frac{1}{2}}$$

$$= \frac{1}{\sqrt{2}}$$

$$\therefore a = \frac{4}{e}$$

$$= \frac{4}{\frac{1}{\sqrt{2}}}$$

$$= 4\sqrt{2} \Rightarrow \boxed{B}$$

since ellipse

Q4. chord of contact:

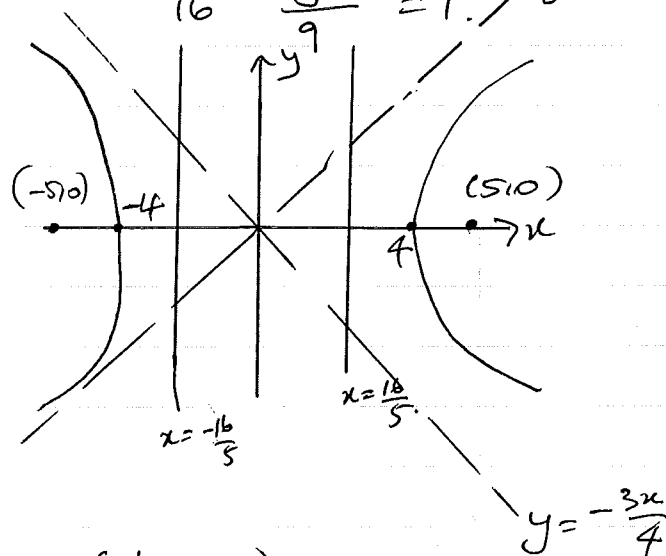
$$\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$$

$a^2 = 10, b^2 = 5$.

$\therefore \boxed{A}$.

SECTION II

S. (a) $\frac{x^2}{16} - \frac{y^2}{9} = 1$ / $y = \frac{3x}{4}$



$S(\pm 4, 0)$

$e = \frac{5}{4}$.

① shape with vertex & focus.

② as above + directrices

③ as above + asymptotes

$$5(b) \quad \frac{x^2}{3} - \frac{y^2}{2} = 1$$

$$a = \sqrt{3}, \quad b = \sqrt{2}$$

$$e = \sqrt{1 + \frac{2}{3}} \\ = \sqrt{\frac{5}{3}} \quad \checkmark$$

$$(i) \quad S = (\pm ae, 0) \\ = (\pm \sqrt{3} \times \frac{\sqrt{5}}{\sqrt{3}}, 0) \\ = (\pm \sqrt{5}, 0) \quad \checkmark$$

- ② correct answer
① focus $(\pm \sqrt{3}e, 0)$

Tangent:

$$(ii) \quad \sqrt{2}x \sec \theta - \sqrt{3}y \tan \theta = \sqrt{6}$$

Normal:

$$\sqrt{3}x \tan \theta + \sqrt{2}y \sec \theta = k.$$

sub (Point P)

$$\therefore k = 3 \tan \theta \sec \theta + 2 \tan \theta \sec \theta$$

$$k = 5 \tan \theta \sec \theta$$

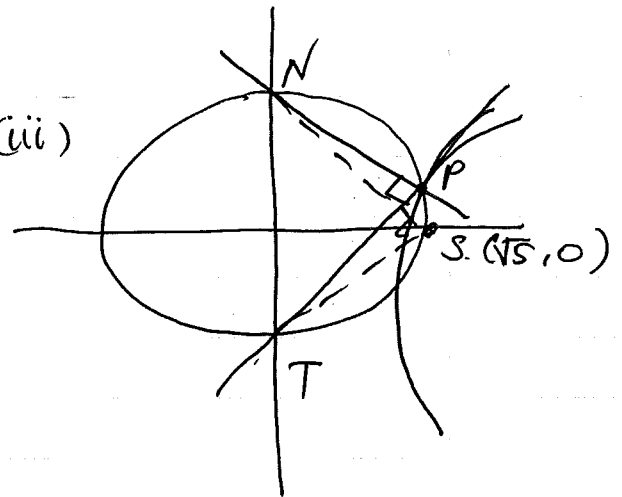
Normal is:

$$\sqrt{3}x \tan \theta + \sqrt{2}y \sec \theta \\ = 5 \tan \theta \sec \theta$$

- ② Full correct equation

- ① only finds k.

(ii)



know $\angle NPT = 90^\circ$
trying to prove $\angle NST = 90^\circ$

$$T(0, -\frac{\sqrt{2}}{\tan \theta}) \quad \checkmark$$

$$N(0, \frac{5 \tan \theta}{\sqrt{2}}) \quad \checkmark$$

$$m_{TS} = -\frac{\sqrt{2}}{\tan \theta} \div \sqrt{5} \\ = -\frac{\sqrt{2}}{\sqrt{5} \tan \theta}$$

$$m_{SN} = \frac{5 \tan \theta}{\sqrt{2}} \div \sqrt{5} \\ = \frac{5 \tan \theta}{\sqrt{10}} \\ = \frac{\sqrt{5} \tan \theta}{\sqrt{2}} \quad \checkmark$$

$$m_{TS} \times m_{SN}$$

$$= -\frac{\sqrt{2}}{\sqrt{5} \tan \theta} \times \frac{\sqrt{5} \tan \theta}{\sqrt{2}}$$

$$= -1$$

$$\therefore \angle NST = 90^\circ \quad \checkmark$$

$$\text{similarly } \angle NS'T = 90^\circ \quad \checkmark$$

$\therefore$ circle passes through foci

Q6(i) By definition:

$$PS = e \cdot PM.$$

$$PM = \frac{a}{e} - a \cos \theta \\ = \frac{a}{e} (1 - e \cos \theta) \checkmark$$

$$\therefore PS = e \times \frac{a}{e} (1 - e \cos \theta) \checkmark \\ = a (1 - e \cos \theta)$$

$$(ii) PS' = a (1 + e \cos \theta) \checkmark$$

(iii) $P(a \cos \theta, b \sin \theta)$

eqn of tangent:

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \checkmark$$

$$\therefore \frac{ax \cos \theta}{a^2} + \frac{by \sin \theta}{b^2} = 1$$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \checkmark$$

$$bx \cos \theta + ay \sin \theta = ab \checkmark$$

$$\therefore bx \cos \theta + ay \sin \theta - ab = 0. \\ \text{as required}$$

(iv) $S(ae, 0)$

$$\text{Tangent: } b \cos \theta x + ay \sin \theta - ab = 0.$$

$$d_{SL} = \frac{|b \cos \theta (ae) - ab|}{\sqrt{(b \cos \theta)^2 + (a \sin \theta)^2}} \\ = \frac{|ab e \cos \theta - ab|}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} \checkmark$$

similarly

$$d_{S'L'} = \frac{|-ab e \cos \theta - ab|}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} \checkmark$$

$$(v) \frac{SL}{S'L'} = \frac{|ab e \cos \theta - ab|}{|-ab e \cos \theta - ab|} \\ = \frac{ab |e \cos \theta - 1|}{-ab |e \cos \theta + 1|} \\ = - \frac{e \cos \theta - 1}{e \cos \theta + 1}.$$

$$\frac{PS}{PS'} = \frac{a (1 - e \cos \theta)}{a (1 + e \cos \theta)} \\ = \frac{1 - e \cos \theta}{1 + e \cos \theta}$$

$$\therefore \frac{SL}{S'L'} = \frac{PS}{PS'} \checkmark$$

Hence $\frac{PL}{PL'} = \frac{PS}{PS'}$

By Pythagoras' theorem

$\therefore \triangle PSL \parallel \triangle PS'L'$

$\therefore \angle LPS = \angle L'PS' \quad \checkmark$

$\therefore \angle NPL = 90^\circ$

$\angle NPS = 90^\circ - \angle LPS$

similarly $\angle NPS' = 90^\circ - \angle L'PS' \quad \checkmark$

$\therefore \angle NPS = \angle NPS'$

$\therefore PN$ bisects $\angle S'PS$

[The tangent at P on the ellipse is equally inclined to the focal chords through P.]

(vi) Since $\triangle PSN \parallel \triangle PS'N \quad \checkmark$
(2 sides in proportion $\checkmark$
about an equal angle).

$\therefore \frac{PS}{NS} = \frac{PS'}{NS'}$

Integration / 28

1. [D]

$$\int_{\sqrt{3}}^2 \sqrt{4-x^2} \, dx$$

$$x = 2 \sin \theta$$

$$dx = 2 \cos \theta \, d\theta$$

$$\text{when } x = 2, \theta = \frac{\pi}{2}$$

$$x = \sqrt{3}, \theta = \frac{\pi}{3}$$

$$\therefore \int_{\sqrt{3}}^2 \sqrt{4-x^2} \, dx$$

$$= 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sqrt{4(1-\sin^2 \theta)} \cdot \cos \theta \, d\theta$$

$$= 4 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 \theta \, d\theta$$

etc.

2. [B]

$$\frac{d}{dx} (x^2 e^x + c)$$

$$= x^2 (e^x) + e^x (2x)$$

$$= e^x x^2 + 2x e^x$$

3. [C]

$$\int \frac{dx}{\sqrt{7-6x-x^2}}$$

$$= \int \frac{dx}{\sqrt{16-(x+3)^2}}$$

$$\times \text{ use } \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$4. \text{ (B). } I_4 = \frac{4-1}{4} I_2$$

$$I_2 = \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$$

$$= \frac{\pi}{4}$$

$$\therefore I_4 = \frac{3}{4} \times \frac{\pi}{4}$$

$$= \frac{3\pi}{16}$$

$$5(a) \int \frac{x^2}{\sqrt{x^3+1}} \, dx$$

$$u = x^3 + 1$$

$$du = 3x^2 \, dx$$

$$= \frac{1}{3} \int \frac{1}{\sqrt{u}} \, du \quad \checkmark$$

$$= \frac{1}{3} [2u^{\frac{1}{2}}] + C$$

$$= \frac{2}{3} \sqrt{x^3+1} + C. \quad \checkmark$$

$$b) \int_0^{\frac{1}{2}} \sin^{-1} x \, dx$$

$$u = \sin^{-1} x \quad v = x$$

$$du = \frac{1}{\sqrt{1-x^2}} \, dx \quad dv = dx.$$

$$\therefore \int_0^{\frac{1}{2}} \sin^{-1} x \, dx \quad \checkmark$$

$$= [x \sin^{-1} x]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} \, dx$$

$$= \frac{1}{2} x \sin^{-1}(\frac{1}{2}) - \int_0^{\frac{1}{2}} \frac{x \, dx}{\sqrt{1-x^2}}$$

$$\text{let } w = 1-x^2$$

$$dw = -2x \, dx$$

$$= \frac{1}{2} \times \frac{\pi}{6} - \int_0^{\frac{3}{4}} \frac{\frac{3}{4}x}{\sqrt{w}} \times \frac{-1}{2x} \, dw \quad \checkmark$$

$$= \frac{\pi}{12} + \frac{1}{2} \int_0^{\frac{3}{4}} w^{-\frac{1}{2}} \, dw$$

$$= \frac{\pi}{12} + [\sqrt{w}]_0^{\frac{3}{4}}$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} \quad \checkmark$$

$$(c) \frac{x^2-4x+2}{(2x+1)(x^2+4)} = \frac{A}{2x+1} + \frac{Bx+C}{x^2+4}$$

$$x^2-4x+2 = A(x^2+4) + (Bx+C)(2x+1)$$

$$\text{when } x=0.$$

$$\boxed{2 = 4A + C}$$

$$x=1 : \boxed{-1 = 5A + 3B + 3C}$$

$$x=-\frac{1}{2} : \boxed{A=1} \quad \checkmark$$

$$\boxed{B=0} \quad \checkmark \quad \boxed{C=-2} \quad \checkmark$$

$$\begin{aligned}
 \text{(ii)} \quad & \int_0^2 \frac{x^2 - 4x + 2}{(2x+1)(x^2+4)} dx \\
 &= \int_0^2 \frac{1}{2x+1} - \frac{2}{x^2+4} dx \\
 &= \left[\frac{1}{2} \ln(2x+1) \right]_0^2 - 2 \left[\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) \right]_0^2 \\
 &= \frac{1}{2} \ln 5 - \tan^{-1}(1) \\
 &= \frac{1}{2} \ln 5 - \frac{\pi}{4}. \\
 &\text{or } \ln \sqrt{5} - \frac{\pi}{4} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 &= \int_{\frac{1}{\sqrt{3}}}^1 \frac{1}{t} + t dt \\
 &= \left[\ln(t) + \frac{t^2}{2} \right]_{\frac{1}{\sqrt{3}}}^1 \\
 &= \ln 1 - \frac{1}{2} - \left(\ln\left(\frac{1}{\sqrt{3}}\right) + \frac{\frac{1}{3}}{2} \right) \\
 &= -\frac{1}{2} + \ln \sqrt{3} + \frac{2}{3} \\
 &= \frac{1}{6} + \ln \sqrt{3} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{6(a)} \quad & t = \tan\left(\frac{\theta}{2}\right) \\
 & \frac{\theta}{2} = \tan^{-1} t \quad \checkmark \\
 & \frac{d\theta}{2} = \frac{1 dt}{1+t^2} \quad \checkmark \\
 & d\theta = \frac{2 dt}{1+t^2}
 \end{aligned}$$

$$\text{Now } \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 2 \operatorname{cosec}^2 \theta \tan\left(\frac{\theta}{2}\right) d\theta$$

$$t_1 = \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$$

$$t_2 = \tan\left(\frac{\pi}{4}\right) = 1$$

$$\begin{aligned}
 \operatorname{cosec}^2 \theta &= \frac{1}{\sin^2 \theta} \\
 &= \frac{(1+t^2)^2}{4t^2}
 \end{aligned}$$

$$\begin{aligned}
 & 2 \int_{\frac{1}{\sqrt{3}}}^1 \frac{(1+t^2)^2}{4t^2} \cdot t \cdot \frac{2 dt}{1+t^2} \\
 &= \int_{\frac{1}{\sqrt{3}}}^1 \frac{1+t^2}{t} dt \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) (i)} \quad & I_{2n+1} = \int_0^1 x^{2n+1} \cdot e^{x^2} dx \\
 & u = x^{2n} \quad v' = x e^{x^2} \\
 & du = 2n x^{2n-1} \quad v = \frac{e^{x^2}}{2}
 \end{aligned}$$

IBP:

$$\begin{aligned}
 & \int_0^1 x^{2n} \cdot x \cdot e^{x^2} dx \quad \checkmark \\
 &= \left[x^{2n} \cdot \frac{e^{x^2}}{2} \right]_0^1 - \int_0^1 \frac{e^{x^2}}{2} \cdot 2n x^{2n-1} dx \quad \textcircled{1} \\
 &= \frac{e}{2} - n \int_0^1 x^{2n-1} \cdot e^{x^2} dx \quad \checkmark \\
 &= \frac{e}{2} - n I_{2n-1} \quad \textcircled{2}
 \end{aligned}$$

$$(ii) I_5 = \frac{e}{2} - 2I_3.$$

$$= \frac{e}{2} - 2\left[\frac{e}{2} - I_1\right].$$

$$= -\frac{e}{2} + 2I_1. \quad \checkmark \textcircled{1}$$

$$I_1 = \int_0^1 x e^{x^2} dx$$

$$= \frac{1}{2} \int_0^1 2x e^{x^2} dx$$

$$= \left. \frac{e^{x^2}}{2} \right|_0^1$$

$$= \frac{1}{2} (e - 1) \quad \checkmark \textcircled{1}$$

$$\therefore I_5 = -\frac{e}{2} + 2\left(\frac{1}{2}(e - 1)\right)$$

$$= -\frac{e}{2} + e - 1$$

$$= \frac{e}{2} - 1 \quad \checkmark \textcircled{1}.$$